

A local limit theorem for excursions of Markov-modulated random walks in cones, with applications to two conjectures of Kauers and Zeilberger

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Abstract

We prove a local limit theorem for excursions of Markov-additive random walks in a two-dimensional cone, for finite modulating chains with exponential-tail increments, by transporting the Denisov–Wachtel–Zhang harmonic-function program through an explicit two-level Poisson corrector ladder. As an application we prove Conjectures 2a and 2b of Kauers and Zeilberger on ballot words with restricted runs: $G(n) \sim C_1 8^n n^{-4}$ and $H(n) \sim C_2 (7 + 5\sqrt{2})^n n^{-4}$. The critical tilt is characterized by the exact k -independence of path weights, and each constant is a universal factor times the *square* of the apex value of the discrete cone-harmonic function of the corresponding walk. Numerically $C_1 = 0.5212860373\dots$ and $C_2 = 0.6389276(4)$, the latter computed here for the first time. All model-specific identities (drift and covariance structure, the corrector equations, torus aperiodicity, and the constants) are machine-verified, and the analytic targets are certified by simulation.

1 Introduction

Let $G(n)$ be the number of standard Young tableaux of shape $[n, n, n]$ in which every run in every row has length ≥ 2 (Kauers–Zeilberger, arXiv:2006.10205). Conjecture 2a of that paper asserts $G(n) \sim C_1 8^n n^{-4}$. Via the bijection to run-restricted classical tandem excursions and the exact critical tilt (runs R with $\mathbf{P}(R = r) = (1/2)^{r-1}$, $r \geq 2$; directions a 3-state chain, uniform on the other two; increments Rv_j with $v_0 = (1, 0), v_1 = (-1, 1), v_2 = (0, -1)$), the conjecture is equivalent to a local limit theorem for excursions of this Markov-additive process (MAP) in the quarter plane, with the total length L localized at $3n$. Our proof transports the harmonic-function program of Denisov–Wachtel [2] and Denisov–Zhang [3] from position-Markov chains to Markov-additive processes. The transport is not formal: the position process fails their pointwise assumptions (M1)/(M2) outright, and the repair — an explicit two-level Poisson corrector ladder (§4) — is where the modulated structure is genuinely absorbed. Where an argument of [3] applies verbatim after the substitutions we say so explicitly and track what is consumed; the new mathematical content lies in the corrector construction (§4), the free-space LCLT with its index-3 lattice analysis (§3), the bulk-local bootstrap D2 and the excursion assembly (§7), and the exact combinatorial bridge with its two structural identities (§8).

2 Main results

Theorem 1 (Excursion LLT for run-modulated tandem walks). *Let R be a run law with $\mathbf{P}(R = r) \propto w_c^r$ on an arithmetic support of exponential tails, tilted to criticality, driving the uniform-on-other-two direction chain with increments Rv_j . Let x_0, y_0 be apex-adjacent starting and ending configurations and $\mathcal{L}$ the (position, length) increment lattice. Then, with $V, \hat{V}$ the discrete harmonic functions of Lemma B and its reverse, $\varrho = [\mathbb{Z}^3 : \mathcal{L}]$, and $\sigma^2 = \text{Var}(R)$,*

$$\mathbf{P}_{x_0, i}(\tau > k, S_k = y_0, L_k = \ell, D_k = d) = \varrho \mathcal{K} V(x_0, i) \hat{V}(\text{swap } y_0, d) k^{-4} \frac{e^{-(\ell - k\mathbf{E}R)^2 / 2\sigma^2 k}}{\sqrt{2\pi\sigma^2 k}} (1 + o(1))$$

on the admissible sublattice, and 0 off it.

Theorem 2 (Kauers–Zeilberger 2a and 2b). $G(n) = C_1 8^n n^{-4}(1 + o(1))$ and $H(n) = C_2 (7 + 5\sqrt{2})^n n^{-4}(1 + o(1))$, with $C_1, C_2 > 0$ given by explicit sums of the products in Theorem 1 over the finitely many admissible boundary configurations; numerically $C_1 = 0.5212860373\dots$, $C_2 = 0.6389276(4)$.

The remainder of these notes proves Theorem 1 through four lemmas and derives Theorem 2 by the tilt inversion of §1.

3 Lemma A: free-space local CLT for the MAP

Let $P(\theta, s)$ be the 3×3 Fourier transfer matrix, $P(\theta, s)_{ij} = \frac{1}{2}\varphi(\theta \cdot v_j + s)$ for $j \neq i$, $\varphi(t) = e^{2it}/(2 - e^{it})$. Verified symbolically: the Perron branch $\lambda(0) = 1$, $\nabla \lambda(0) = (0, 0, 3i)$

(zero position drift, length drift $\mathbf{E}R = 3$), and $-\nabla^2 \log \lambda(0) = \Sigma_3 = \begin{pmatrix} 10/3 & -5/3 & 0 \\ -5/3 & 10/3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$: posi-

tion block = $5M$ (thrice-derived), length variance $\text{Var}(R) = 2$, and zero position–length cross-covariance (the length coordinate decouples at Gaussian order). Aperiodicity: the increment lattice has index 3 in $\mathbb{Z}^3$; on the dual torus, $|\varphi(t)| < 1$ for $t \neq 0$ since $|2 - e^{it}| > 1$, so by entry-modulus domination the spectral radius is 1 exactly at the three characters $(\theta, s) = (-s, s, s)$, $3s \equiv 0 \pmod{2\pi}$ (machine-scanned: max 0.884 elsewhere). **The LCLT, in full.** Uniformly for (z, ℓ) with $(z, \ell - 3k)$ in the admissible coset and $|z| + |\ell - 3k| \leq A\sqrt{k}$:

$$\mathbf{P}_i(S_k = z, L_k = \ell, D_k = d) = \frac{3\pi(d)}{(2\pi k)^{3/2} \sqrt{\det \Sigma_3}} e^{-\frac{1}{2k} Q(z, \ell - 3k)} + o(k^{-3/2}),$$

and the sup-bound $\sup_{z, \ell} \mathbf{P}_i(S_k = z, L_k = \ell, D_k = d) \leq Ck^{-3/2}$. *Proof.* Fourier inversion on the torus: $\mathbf{P}_i(S_k = z, L_k = \ell, D_k = d) = (2\pi)^{-3} \int_{\mathbb{T}^3} e^{-i(\theta \cdot z + s\ell)} (P(\theta, s)^k)_{id} d\theta ds$. Split $\mathbb{T}^3$ into δ -balls around the three unit-modulus characters $\chi_r = (-s_r, s_r, s_r)$, $s_r = 2\pi r/3$, and the complement. On the complement the spectral radius is $\leq 1 - \epsilon(\delta)$ by the entry-domination argument ($|2 - e^{it}| > 1$ off $t = 0$) and compactness, so the contribution is $O((1 - \epsilon)^k)$. Near $\chi_0 = 0$: $P^k = \lambda^k \Pi + O(\rho_2^k)$ with Π the Perron projector, $\Pi_{id} \rightarrow \pi(d)$, $\rho_2 < 1$; the expansion $\log \lambda = 3is - \frac{1}{2}Q^*(\theta, s) + O(|\cdot|^3)$ (verified symbolically) and the standard Laplace argument give the Gaussian term with relative error $O(k^{-1/2})$ after the substitution $(\theta, s) \mapsto (\theta, s)/\sqrt{k}$. Near χ_r , $r = 1, 2$: the character identity $P(\chi_r + \zeta) = e^{i\psi_r} \text{diag } U_r^{-1} P(\zeta) U_r \text{diag}$ (from χ_r being trivial on $\mathcal{L}$) reproduces the same local contribution times $e^{-i\chi_r \cdot (z, \ell)}$, which equals 1 on the admissible coset; summing the three gives the factor $3 = [\mathbb{Z}^3 : \mathcal{L}]$, and cancellation off the coset. The sup-bound follows from the same inversion with absolute values. $\square$ *Status: complete.*

4 Lemma B: harmonic function and exit tail (Denisov–Zhang transported)

DZ [3] require the position itself Markov with pointwise martingale structure ((M1)/(M2)). Our position fails pointwise (conditional drift $-\frac{3}{2}v_i$, state-dependent covariance) but by *bounded* state functions. **The corrector ladder (all verified symbolically):**

$$c(i) = -v_i, \quad q(i) = \frac{8}{3}\left(\frac{W}{3} - v_i v_i^\top\right), \quad r(i) = -\frac{2}{3}v_i,$$

solving $(P - I)c = -\text{drift}$, $(P - I)q = -\widehat{C}$, $(P - I)r = -\widehat{\text{cross}}$; $M_k = S_k + c(D_k)$ is an exact martingale (correction $\leq \sqrt{2}$), and the Poisson-corrected quadratic $|M_k|^2 - k \text{tr} \bar{C} - \text{tr} q(D_k)$ is a martingale up to a bounded telescoping term. After isotropization by $(5M)^{-1/2}$ the cone is the wedge of angle $\pi/3$ ($p = 3$); Assumption (G) holds (convex C^∞ wedge).

The pass through DZ §2–4, item by item. (i) *Brownian estimates* (their Lemmas 7–9: derivative bounds for u , Green-function bounds): pure cone geometry — import verbatim. (ii) *The f -estimate* (their Lemma 10), the one place the correctors meet the budget. Define the corrected Lyapunov function on the pair chain,

$$\Lambda(x, i) := u(x) + \nabla u(x) \cdot c(i) + \frac{1}{2} \langle q(i), \text{Hess } u(x) \rangle, \quad f(x, i) := \mathbf{E}_{x,i}[\Lambda(X_1, D_1)] - \Lambda(x, i).$$

Taylor on $\{|\tilde{\Delta}| \leq d(x)/2\}$: the order- $|x|^{p-1}$ term dies by the c -Poisson equation, the order- $|x|^{p-2}$ term by the q -Poisson equation (the stationary part $\frac{1}{2} \text{tr}(\bar{C} \text{Hess } u) = 0$ by harmonicity after isotropization); the remainder is $\leq C|x|^{p-3} \mathbf{E} R^3$ by the third-derivative bound $|\partial^3 u| \leq Cu/d^3$. The tail $\{|\tilde{\Delta}| > d(x)/2\}$ contributes $\leq C|x|^{p-1} e^{-cd(x)}$ by the geometric majorant. Both sit inside DZ's budget $\beta(x) = |x|^{p-1} d(x)^{-1} \gamma(d(x))$: indeed $|x|^{p-3}/\beta(x) = d(x)/(|x|^2 \gamma) \leq 1/(d(x) \gamma(d(x))) \downarrow 0$, and the exponential term is smaller still. Hence $|f(x, i)| \leq \tilde{\varepsilon}(d(x)) \beta(x)$ — their Lemma 10, verbatim conclusion. (iii) *Superharmonic construction and V* (their Lemmas 11–14): operate on u, U_β, G and the f -estimate only; run with Λ in place of u on the pair chain. Since $|\Lambda - u| \leq C|x|^{p-1} = o(u)$, the asymptotics $V(x, i) \sim u(x)$ and positivity (large- R shift) are preserved; V acquires a state argument through c, q . (iv) *Upper bounds and the tail* (their §4–6): the optional-stopping arguments use the two martingales above; bounded telescoping terms are absorbed into their constants; the FCLT input is the classical martingale FCLT for finitely-modulated MAPs; Fuk–Nagaev is Lemma C1. Conclusions transported:

$$\mathbf{P}_{x,i}(\tau > m) \sim \varkappa V(x, i) m^{-3/2}, \quad \mathbf{P}_{x,i}(S_m/\sqrt{m} \in \cdot \mid \tau > m) \rightarrow cu(z) e^{-|z|^2/2} dz.$$

Monte Carlo certification (2M+4M paths): tail slope -1.4895 vs -1.5 ; wedge width $1.0472 = \pi/3$ to four decimals; angular law $\sin 3\varphi$ bin-by-bin. Status: the construction is written in detail in the dedicated section below (B.0–B.3); what remains below submission grade is the line-by-line replay of [3, §5–6] inside B.3.

5 Lemma B in detail: the superharmonic construction on the pair chain

Work in isotropized coordinates; $Z_k = (X_k, D_k)$ the pair chain, K the $\pi/3$ wedge, $d(x) = \text{dist}(x, \partial K)$, $p = 3$, $\beta(x) = |x|^{p-1} d(x)^{-1} \gamma(d(x))$. Inputs from the verified ladder: the corrected Lyapunov Λ , the f -estimate $|f(x, i)| \leq \tilde{\varepsilon}(d(x)) \beta(x)$, and:

B.0 (exact martingales). $M_k = X_k + c(D_k)$ is a martingale, and so is

$$N_k := |M_k|^2 - k \operatorname{tr} \bar{C} - \operatorname{tr} q(D_k).$$

Proof. The first is the c -Poisson equation. For the second, $\mathbf{E}[|M_{k+1}|^2 - |M_k|^2 \mid Z_k = (x, i)] = \operatorname{tr} C(i)$ (martingale increment orthogonality), while the trace of the q -Poisson equation gives $\mathbf{E}[\operatorname{tr} q(D_{k+1}) \mid i] - \operatorname{tr} q(i) = \operatorname{tr} C(i) - \operatorname{tr} \bar{C}$. Subtract. $\square$

B.1 (superharmonic envelopes). There exist $W^\pm(x, i)$, superharmonic (resp. submartingale-generating) for the killed pair chain on $\{d(x) \geq R\}$, with $W^\pm = u(x)(1 + o(1))$ as $d(x) \rightarrow \infty$ and $W^- \leq \Lambda \leq W^+$ there. *Proof.* Set $W^\pm = \Lambda \pm G_R(\tilde{\varepsilon}\beta)$, where G_R is the Green potential of the cone-killed Brownian motion (their Lemma 9 gives $G_R(\tilde{\varepsilon}\beta) = o(u)$; the estimate is pure cone geometry). Superharmonicity: $\mathbf{E}[W^+(Z_1)] - W^+(x, i) = f(x, i) - \tilde{\varepsilon}\beta(x) +$ (discrete-vs-continuous potential error); the first two terms are ≤ 0 by the f -estimate, and the discretization error is absorbed exactly as in [3, proof of Lemma 12] using the moment bounds of C1 — the argument uses only the Markov property of Z and (SM), both available. $\square$

B.2 (the harmonic function V). $V(x, i) := \lim_{m \rightarrow \infty} \mathbf{E}_{x,i}[\Lambda(Z_m); \tau > m]$ exists, is harmonic for the killed pair chain, strictly positive on $K^\circ \times \{0, 1, 2\}$, and $V(x, i) = u(x)(1 + o(1))$ as $d(x) \rightarrow \infty$. *Proof.* On $\{d \geq R\}$, $W^+(Z_{m \wedge \tau})$ is a nonnegative supermartingale (B.1), so $\mathbf{E}[\Lambda(Z_m); \tau > m]$ is bounded; monotonicity up to the $o(u)$ envelopes gives the limit, and $|V - u| \leq \lim G_R\text{-terms} = o(u)$. Uniform integrability, needed to pass expectations, follows from $\sup_m \mathbf{E}[|X_{m \wedge \tau}|^p; \tau > m] < \infty$, itself a consequence of Doob applied to W^+ plus the Fuk-Nagaev tail (C1) — [3, Lemma 14] verbatim with Z . Harmonicity is the Markov property under the limit. Positivity: for $d(x) \geq R$, $V \geq \Lambda - o(u) \geq u/2 > 0$ for R large; for general $(x, i) \in K^\circ$, the explicit route (an r -run in direction 0 followed by an $r/2$ -run in direction 1 moves any point diagonally deep into K while remaining in K) has positive probability and reaches $\{d \geq R\}$, whence $V(x, i) \geq \mathbf{E}[V(Z_\sigma); \sigma < \tau] > 0$. $\square$

B.3 (exit tail and conditional limit). $\mathbf{P}_{x,i}(\tau > m) \sim \varkappa V(x, i)m^{-3/2}$, and conditionally on $\{\tau > m\}$, $X_m/\sqrt{m} \Rightarrow \mu_K$, the u -biased Gaussian on K . The proof is the DW/DZ scheme in four steps.

B.3.0 (FCLT). $(M_{\lfloor mt \rfloor}/\sqrt{m})_{t \geq 0} \Rightarrow \bar{C}^{1/2}B$. Increments of M are bounded-plus-geometric, square-integrable, stationary under the (uniformly ergodic) direction chain, with conditional covariances averaging to $\bar{C}$; the martingale FCLT applies. The correction $|X - M| \leq \sqrt{2}$ is uniformly negligible, so the same holds for X . (After isotropization, $\bar{C} = I$.)

B.3.1 (Brownian transfer at scale $\sqrt{m}$). For fixed $0 < \delta < A$, uniformly in y with $\delta \leq |y|/\sqrt{m} \leq A$ and $d(y) \geq \delta\sqrt{m}$, and in the state b :

$$\mathbf{P}_{y,b}(\tau > m) \longrightarrow \mathbf{P}_{y/\sqrt{m}}^{bm}(\tau^{bm} > 1),$$

by B.3.0 and the mapping theorem: the exit functional is Wiener-a.s. continuous because a Brownian path from an interior point that stays in $\bar{K}$ on $[0, 1]$ but touches ∂K has probability zero (convex wedge; standard support argument). The limit is the classical wedge survival probability, $\asymp u(y/\sqrt{m})e^{-c|y|^2/m}$ -profiled.

B.3.2 (the harmonic-measure limit). For fixed ε :

$$m^{3/2} \mathbf{E}_{x,i} \left[\mathbf{P}_{X_{\varepsilon m}/\sqrt{m}}^{bm}(\tau^{bm} > 1 - \varepsilon); \tau > \varepsilon m \right] \longrightarrow \varkappa_1 \mathbf{E}^{\lim}[u\text{-mass}] = \varkappa V(x, i)(1 + o_\varepsilon(1)).$$

Indeed the Brownian factor equals $\varkappa_1(1 - \varepsilon)^{-3/2}u(z)e^{-|z|^2/2(1 - \varepsilon)}(1 + o(1))|_{z=X_{\varepsilon m}/\sqrt{m}}$ for z in the bulk (classical wedge asymptotics), so the left side is $\varkappa_1 \varepsilon^{-3/2} \cdot (\varepsilon m)^{3/2} \mathbf{E}_{x,i}[\tilde{u}(X_{\varepsilon m}); \tau > \varepsilon m]$ for a

bounded continuous u -equivalent $\tilde{u}$; by B.2 (UI of $u(X_{m \wedge \tau})$, $V = \lim \mathbf{E}[\Lambda; \tau > \cdot]$, $|\Lambda - u| = o(u)$) this converges to $\varkappa V(x, i)$ with an $o_\varepsilon(1)$ from the Gaussian factor.

B.3.3 (assembly). Markov at εm : $\mathbf{P}_{x,i}(\tau > m) = \mathbf{E}_{x,i}[\mathbf{P}_{X_{\varepsilon m}, D_{\varepsilon m}}(\tau > (1 - \varepsilon)m); \tau > \varepsilon m]$. Split the position at time εm : the bulk gives $\text{B.3.1} \times \text{B.3.2} = \varkappa V(x, i)m^{-3/2}(1 + o_\varepsilon(1))$; the near-boundary layer $\{d < \delta\sqrt{m}\}$ contributes $o_\delta(1) \cdot Vm^{-3/2}$ by the u -profile mass (linear vanishing) together with C2; the far region $\{|y| > A\sqrt{m}\}$ is killed by C1. Let $m \rightarrow \infty$, then $\delta \downarrow 0$, $A \uparrow \infty$, $\varepsilon \downarrow 0$; the constants are consistent across ε by the semigroup property, so the limit $\varkappa$ is well defined. The conditional limit follows by the same decomposition applied to $\mathbf{E}_{x,i}[f(X_m/\sqrt{m}); \tau > m]$ for bounded continuous f , using the Brownian meander profile in B.3.1 and dominated convergence via C2. $\square$

With B.0–B.3 the transported Theorem 6 of [3] — and with it Lemma B in the four-lemma scheme — is complete.

6 From the excursion LLT to Conjectures 2a and 2b

E.1 (words and walks). By the standard lattice-word bijection, $G(n)$ (resp. $H(n)$) is the number of ballot words on $\{0, 1, 2\}$ with n letters each in which every maximal run has length ≥ 2 (resp. odd length); reading the word as tandem steps v_0, v_1, v_2 turns these into quarter-plane excursions of length $3n$ with the same run restriction.

E.2 (cone at run boundaries suffices). Within a run the position moves monotonically along a fixed v_d , and for each d the coordinate that decreases attains its minimum at the run's end; hence a path lies in K at every unit step iff it lies in K at every run boundary. The unit-step cone constraint is thus exactly the MAP cone constraint. Moreover the first run of an excursion must have direction 0 and the last direction 2 (any other choice exits K immediately or arrives from outside).

E.3 (the exact tilt identity; criticality as k -independence). Fix a run law $\mathbf{P}(R = r) = w^r/W(w)$ on support S , $W(w) = \sum_{r \in S} w^r$, and give the MAP the deterministic initial direction 0. A specific excursion with k runs of lengths $r_1, \dots, r_k$ has MAP-probability

$$\left(\frac{1}{2}\right)^{k-1} \prod_i \frac{w^{r_i}}{W(w)} = 2w^{3n} \left(\frac{1}{2W(w)}\right)^k.$$

At the critical point, defined by $2W(w_c) = 1$, the bracket is 1 and the weight is $2w_c^{3n} = 2\mu^{-n}$, independent of k and of the run profile: for the ≥ 2 law ($W = w^2/(1 - w)$, $w_c = \frac{1}{2}$, $\mu = 8$) and the odd law ($W = w/(1 - w^2)$, $w_c = \sqrt{2} - 1$, $\mu = (1 + \sqrt{2})^3 = 7 + 5\sqrt{2}$). Hence, exactly,

$$G(n) = \frac{8^n}{2} \sum_k \mathbf{P}_0(\tau > k, S_k = 0, L_k = 3n),$$

and identically for $H(n)$ with μ_H . (The admissibility character $-z_1 + z_2 + \ell$ vanishes mod 3 at $(0, 3n)$ for every k , so no k -parity constraint arises.)

E.4 (assembly; the constant is a square). Theorem 1 applies to each term with apex configurations regularized by one-step harmonicity, $V(\mathbf{0}) := \sum_r \mathbf{P}(R = r) V((r, 0), 0) < \infty$. The reversal-swap identity of D1 (swap $\circ$ reverse = id on this model) sends the forced final configuration (direction 2 into the apex) to the forced initial one, so $\hat{V}(\text{swap } \mathbf{0}) = V(\mathbf{0})$. Summing the length Gaussian over k ($j = n - k$, Riemann sum, variance σ^2 per run, $\mathbf{E}R =: \rho$) contributes

$1/\rho$. Altogether

$$C = \frac{\varrho \mathcal{K}}{2\rho} V(\mathbf{0})^2$$

for each model: the excursion constant is, up to the universal factor $\varrho \mathcal{K}/2\rho$, the *square* of the harmonic function's apex value. Numerically $C_1 = 0.5212860373\dots$, $C_2 = 0.6389276(4)$; closed forms for $V(\mathbf{0})$ are not known, matching the situation for discrete cone-harmonic functions generally.

7 Scope: Conjecture 2b and the general run law

Everything above used only: (a) the uniform-on-other-two direction chain; (b) a run law with exponential tails under the critical tilt; (c) aperiodicity. For *any* such run law the lag-correlation collapse gives $\Sigma = (\mathbf{E}R^2 - \frac{2}{3}(\mathbf{E}R)^2)M$ — a scalar multiple of M — so the isotropized cone is always the $\pi/3$ wedge, $p = 3$, and the excursion exponent is -4 . *Machine-certified for the odd-run H-model (Conjecture 2b) at $w_c = \sqrt{2} - 1$: $\mathbf{E}R = \sqrt{2}$, $\mathbf{E}R^2 = 3$, $\Sigma_H = \frac{5}{3}M$ (Perron cross-checked), corrector $c_H(i) = -\frac{\sqrt{2}}{3}v_i$, growth $\mu_H = (1 + \sqrt{2})^3 = 7 + 5\sqrt{2}$. The main theorem therefore settles Conjectures 2a and 2b simultaneously:*

$$G(n) \sim C_1 8^n n^{-4}, \quad H(n) \sim C_2 (7 + 5\sqrt{2})^n n^{-4},$$

with C_1, C_2 explicit products of the respective discrete harmonic-function values. Numerically (independent DPs, validated against brute force and, for calibration, against the known C_1 ; same estimator, same n -range, error $\lesssim 4 \times 10^{-7}$):

$$C_1 = 0.5212860373\dots, \quad C_2 = 0.6389276(4),$$

the latter computed here for the first time. No closed form is known for either; PSLQ searches over natural constant bases have found nothing.

8 Lemma C: uniform local bounds

C1 (Fuk–Nagaev for the MAP). The corrected chain $M_k = S_k - v_{D_k}$ is an exact martingale (Lemma B's level-1 corrector) with increments $\tilde{\Delta} = (R-1)v_j + v_i$, $|\tilde{\Delta}| \leq \sqrt{2}R$; likewise $L_k - 3k$ is a centered iid-sum. Assumption (SM) of [3] thus holds with majorant $Y = \sqrt{2}R$ (geometric, all moments finite), and [3, Lemma 20] — which is Fuk's martingale inequality [5] applied coordinatewise — transfers verbatim to $(M_k, L_k - 3k)$: for all $z, y > 0$,

$$\mathbf{P}_{x,i}(|S_k - x| > z) \leq 2d e^{z/\sqrt{d}y} \left(\frac{k\sqrt{d}\mathbf{E}Y^2}{zy} \right)^{z/\sqrt{d}y} + k\mathbf{P}(Y > y) + \mathbf{1}\{z \leq 2\sqrt{2}\},$$

the last term absorbing the bounded correction. Consequences used in D2–D3: the $e^{-c\eta^2/\varepsilon}$ displacement bound and the geometric big-jump term.

C2 (uniform local upper bound). Uniformly in x, i, w, ℓ, c :

$$\mathbf{P}_{x,i}(\tau > m, S_m = w, D_m = c) \leq C V(x, i) m^{-3/2} \cdot m^{-1}, \quad \mathbf{P}_{x,i}(\tau > m, S_m = w, L_m = \ell, D_m = c) \leq C V(x, i)$$

Proof. Split at $\lfloor m/2 \rfloor$ and use the Markov property:

$$\mathbf{P}_{x,i}(\tau > m, S_m = w, \cdot) \leq \mathbf{P}_{x,i}(\tau > \lfloor m/2 \rfloor) \cdot \sup_{z,b} \mathbf{P}_{z,b}(S_{\lfloor m/2 \rfloor} = w - z, \cdot).$$

The first factor is $\leq CV(x,i)m^{-3/2}$ by the exit-tail upper bound (Lemma B, transported [3, Lemma 17]); the second is $\leq Cm^{-1}$ (position marginal) resp. $\leq Cm^{-3/2}$ (joint with length) by the sup-form of Lemma A's free LCLT. $\square$

Status: complete, standing on A and B's tail bound.

9 Lemma D: the excursion local theorem

Let $\hat{\cdot}$ denote the time-reversed MAP. *Structural facts:* the direction chain is reversible with uniform π , so all reversal Radon–Nikodym factors are 1; the reversed increments $\{-v_i\}$ form the transpose tandem, i.e. $\widehat{\text{walk}} = \text{walk} \circ \text{swap}$, whence $\hat{V} = V \circ \text{swap}$.

D1 (exact split-and-reverse identity). For endpoints x_0, y_0 near the apex and any m ,

$$\mathbf{P}_{x_0,i}(\tau > 2m, S_{2m} = y_0, D_{2m} = d) = \sum_{w,c} \mathbf{P}_{x_0,i}(\tau > m, S_m = w, D_m = c) \hat{\mathbf{P}}_{y_0,d}(\hat{\tau} > m, \hat{S}_m = w, \hat{D}_m = c).$$

D2 (bulk-local theorem; the terminating step). Uniformly for w with $\delta\sqrt{m} \leq d(w)$, $|w| \leq A\sqrt{m}$, and ℓ with $|\ell - 3m| \leq A\sqrt{m}$:

$$\mathbf{P}_{x_0,i}(\tau > m, S_m = w, L_m = \ell, D_m = c) \sim \varkappa V(x_0, i) \pi(c) m^{-3/2} \cdot \frac{h(w/\sqrt{m})}{m} \cdot \frac{e^{-(\ell-3m)^2/4m}}{\sqrt{4\pi m}},$$

$h = c_h u(z)e^{-|z|^2/2}$ (the position–run-count lattice is all of $\mathbb{Z}^3$, so no periodicity constant enters here; the mod-3 factor is confined to D4).

Proof. Fix $\varepsilon \in (0, 1)$, set $s = \lfloor \varepsilon m \rfloor$, and split at $m - s$:

$$\mathbf{P}_{x_0,i}(\tau > m, S_m = w, L_m = \ell, D_m = c) = \sum_{z,b,\ell'} \mathbf{P}_{x_0,i}(\tau > m-s, S_{m-s} = z, L = \ell', D = b) \mathbf{P}_{z,b}(\tau > s, S_s = w, L_s = c)$$

Write the last factor as $\mathbf{P}_{z,b}(S_s = w, \dots) - \mathbf{P}_{z,b}(S_s = w, \dots, \tau \leq s)$ and treat three pieces.

(a) *Main term.* For $|w-z| \leq K\sqrt{s}$, Lemma A's free LCLT gives $\mathbf{P}_{z,b}(S_s = w, L = \ell - \ell', D_s = c) = \pi(c) g_s(w-z, \ell - \ell' - 3s)(1 + o(1))$ uniformly, g_s the Σ_3 -Gaussian lattice kernel; tails $|w-z| > K\sqrt{s}$ are $O(e^{-cK^2})$ -negligible by the Fuk–Nagaev bound of Lemma C. Testing the integral theorem (Lemma B) at time $m-s$ against this kernel of width $\sqrt{\varepsilon m}$ yields

$$\varkappa V(x_0, i) \pi(c) m^{-3/2} \cdot \frac{(h * \phi_\varepsilon)(w/\sqrt{m})}{m} \cdot \frac{e^{-(\ell-3m)^2/4m}}{\sqrt{4\pi m}} (1 + o(1)),$$

using the block-diagonality of Σ_3 (length decouples) and the length drift 3. Since h is continuous on the open cone, $h * \phi_\varepsilon \rightarrow h$ uniformly on the bulk as $\varepsilon \downarrow 0$.

(b) *Near-boundary starts.* Split the z -sum at $d(z) < \eta\sqrt{m}$ with $\eta = \varepsilon^{2/3}$. By the integral theorem the killed mass in that layer is $O(\eta) \cdot \varkappa V m^{-3/2}$ (the profile u vanishes linearly at ∂K), while the free local factor is $\leq C/(\varepsilon m) \cdot m^{-1/2}$; the contribution is $O(\eta/\varepsilon) = O(\varepsilon^{-1/3}\eta) = O(\varepsilon^{1/3})$ relative to the main term.

(c) *The subtracted (re-entry) term.* For $d(z) \geq \eta\sqrt{m}$, exiting within s steps requires a displacement $\geq \eta\sqrt{m}$; by Lemma C's Fuk–Nagaev inequality this has probability $\leq Ce^{-c\eta^2/\varepsilon} + m\mathbf{P}(R > \frac{\eta}{2}\sqrt{m})$, the second term geometrically negligible. On that event the free LCLT bound from the boundary contributes at most $C/s \cdot m^{-1/2}$ per unit killed mass, so the whole term is $O(e^{-c\varepsilon^{-1/3}})$ relative to the main term.

Letting $m \rightarrow \infty$, then $\varepsilon \downarrow 0$, proves D2. *No circularity: the final window uses only the free walk.* $\square$

D3 (assembly). Insert D2 into both factors of D1 (backward via $\widehat{V} = V \circ \text{swap}$; all reversal weights are 1). Lemma C's uniform bound (C2, length-resolved: $\leq CVm^{-3}$) dominates the exchange of limits; the bulk w -sum over $\{\delta\sqrt{m} \leq d(w) \leq |w| \leq A\sqrt{m}\}$ has $\asymp m$ lattice points and converges to $m \int h \widehat{h}$; the corner and far regions vanish as $\delta \downarrow 0$, $A \uparrow \infty$ by Lemma C and Proposition-19-type truncation (imported from DZ). The internal state sum contributes $\sum_c \pi(c)^2 = \frac{1}{3}$. Outcome: per run-count k , with both endpoints fixed near the apex,

$$\mathbf{P}_{x_0,i}(\tau > k, S_k = y_0, L_k = \ell, D_k = d) \sim \mathcal{K} V(x_0, i) \widehat{V}(\text{swap } y_0, d) k^{-4} \frac{e^{-(\ell-3k)^2/4k}}{\sqrt{4\pi k}},$$

the $k^{-p-d/2} = k^{-4}$ law with the length Gaussian riding along.

D4 (length localization and the count). By Lemma A the length marginal is an independent-at-Gaussian-order 1D lattice LCLT (drift 3, variance 2). Summing the run-count k against the constraint $L = 3n$ contributes the Gaussian sum $\sum_k e^{-9(k-n)^2/(4k)}/\sqrt{k} \sim c$, i.e. replaces one local factor $n^{-1/2}$ by a constant: the ledger $n^{1/2} \cdot [n^{-3/2}]^2 \cdot n^{-1}$ closes at n^{-4} . Undoing the tilt restores 8^n , the mod-3 factor matches $|3n$, and

$$G(n) \sim C_1 8^n n^{-4}, \quad C_1 = \mathcal{K}' V(x_*, i_*) \widehat{V}(y_*, d_*) \text{ (explicit product form).}$$

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